AI-Powered Zero-Day Attack Detection Platform — Evaluation Report

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1. Dataset Overview

Dataset: Sample IDS Dataset

Known classes (used for training): benign, probe, r2l, u2r, worm

Unknown / held-out classes (zero-day simulation): dos

2. Model Comparison

All models were trained on identical known-class splits and evaluated with the same held-out unknown classes to ensure a fair comparison.

Model	Acc.	Prec.	Recall	F1	MCC	ROC-AUC	FPR	Unk. Recall	Train (s)	Infer (ms)
cnn	0.896	0.886	0.815	0.842	0.774	0.971	0.032	0.948	1.6	2.03
cnn	0.896	0.886	0.815	0.842	0.774	0.971	0.032	0.948	1.1	1.98

3. Key Finding

cnn achieved the highest unknown-attack recall (0.948), making it the strongest candidate for zero-day detection among the architectures compared.